

“Multifunctional Smart Stick for Visually Impaired with Obstacle Detection”

A MINIPROJECT REPORT

Submitted

In the partial fulfilment of the requirements for
the award of the degree of

BACHELOR OF TECHNOLOGY IN ELECTRONICS AND COMMUNICATION ENGINEERING

By

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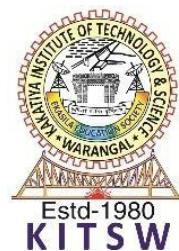
B23EC084

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2025-2026

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CERTIFICATE

This is to certify that Mini Project report entitled “**MULTIFUNCTIONAL SMART STICK FOR VISUALLY IMPAIRED WITH OBSTACLE DETECTION**” embodies the original work done by **GUNDALA ANJALI** bearing Roll Number **B23EC084** studying **VI** Semester in partial fulfilment of the requirement for the award of degree of the **Bachelor of Technology in Electronics & Communication Engineering** from **KAKATIYA INSTITUTE OF TECHNOLOGY & SCIENCE, WARANGAL** during the academic year 2025-2026.

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DECLARATION

I declare that the Mini Project report is original and has been carried out in the Department of Electronics & Communication Engineering, Kakatiya Institute of Technology and Science, Warangal, Telangana, and to the best of my knowledge it has been not submitted elsewhere for any degree.

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ABSTRACT

Visual impairment poses serious challenges for millions of people globally, restricting their safe movement. Traditional mobility tools, including the white cane, help users move freely but have many drawbacks since they allow detecting only very close obstacles and cannot determine whether there are water, pits, or moving objects around. In this case, the visually impaired person may face danger while moving through unfamiliar surroundings.

The current project represents the development of a new device – a Multifunctional Smart Stick for Visually Impaired with Obstacle Detection. The system is created based on an Arduino microcontroller equipped with several additional components. An ultrasonic sensor helps recognize all kinds of obstacles within a certain radius thanks to measuring the distance between the user and these obstacles. Also, a water sensor is added to be able to determine when the floor is wet or when there is a pit nearby.

When a dangerous situation appears, it is instantly communicated to the visually impaired person thanks to the presence of a buzzer or vibration motor in the stick. Moreover, the smart stick contains a GPS module for determining the exact location of the user and sending SMS in case of emergencies thanks to the GSM module installed.

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1. INTRODUCTION

Visual impairment is one of the biggest challenges faced by millions of people across the world on a day-to-day basis. According to a report by the World Health Organization, a significant number of people are either partially blind or completely blind, depending on others for safe passage. The conventional white cane has been the primary assistive tool for the visually impaired for decades. However, its limitations in detecting water, pits, dynamic obstacles, and providing emergency communication have led researchers to develop smarter alternatives.

This project proposes a Multifunctional Smart Stick for visually impaired individuals that integrates several advanced sensing technologies. The proposed system uses an ultrasonic sensor to detect barriers in front of the user and provides real-time feedback via buzzer/vibration motor and voice module alerts. Additionally, a water sensor is incorporated to detect water and pits on the ground, and a GPS and GSM module is included for emergency communication by sending the user's location to a pre-saved contact.

The system is designed to be easy to use, cost-effective, portable, and power-efficient. It greatly improves the safety, confidence, and level of independence of visually impaired users. The project is implemented using the Arduino UNO microcontroller as the central processing unit, which interfaces with all the sensors and output modules to perform the required functions.

1.1 Overview

The Multifunctional Smart Stick is an autonomous assistive device designed to provide a comprehensive safety solution for visually impaired individuals. Unlike traditional white canes that only detect close physical obstacles, this smart stick continuously monitors the environment using multiple sensors to detect static and dynamic obstacles, water, and uneven surfaces.

The system combines both hardware and software components. The hardware handles sensing, actuation, and communication, while the programmed logic controls how the system reacts to different detected conditions. The Arduino UNO microcontroller receives input from all sensors and processes the data in real time to trigger appropriate alerts. The integration of GPS and GSM modules further enhances the system by enabling emergency location-sharing functionality.

1.2 Importance

According to global health statistics, over 250 million people suffer from vision impairment worldwide. The lack of affordable and comprehensive assistive technology means that a majority of these individuals face daily risks. A multifunctional smart stick addresses these risks by providing obstacle detection, hazard alerts, and emergency communication in a single portable device.

Beyond personal safety, such devices contribute to the social independence of visually impaired individuals, enabling them to navigate public spaces like roads, markets, railway stations, and offices without requiring constant human assistance. The project also serves an important educational role by demonstrating how low-cost embedded systems and sensor technologies can be combined to create high-impact assistive solutions.

2. LITERATURE REVIEW

In recent years, assistive technologies for visually impaired individuals have been an active area of research. Several studies have proposed smart stick systems with varying levels of functionality and complexity. Early systems primarily focused on basic obstacle detection using infrared or ultrasonic sensors, while more recent works have incorporated GPS, GSM, voice guidance, and even AI-based technologies.

Kunta, Vanitha et al. (2020) proposed a multifunctional blind stick incorporating ultrasonic obstacle detection along with water and pit sensing features. Their system demonstrated significant improvement over traditional white canes, particularly in the detection of ground-level hazards. The work highlighted the importance of multi-sensor integration in smart assistive devices.

Chen, Liang-Bi et al. (2019) implemented an intelligent assistance system for visually impaired individuals that utilized computer vision and embedded systems. Their approach demonstrated the feasibility of real-time object detection and voice feedback for autonomous navigation, though the cost and computational requirements were considerably higher than simpler sensor-based systems.

Various other researchers have explored the use of GPS and GSM modules for emergency location tracking in smart stick designs. These systems enable the user to send their precise location to a trusted contact in case of emergencies, which is a critical feature for enhancing user safety. The current project builds on these foundational works by integrating obstacle detection, hazard sensing, and emergency communication into a single compact, low-cost, and user-friendly device.

Overall, the literature confirms that multifunctional smart sticks represent a significant advancement over conventional assistive tools. This project synthesizes the key learnings from previous studies and implements them using affordable and readily available components centered around the Arduino platform.

3. OBJECTIVES

The primary objective of this project is to design and develop a multifunctional smart stick for visually impaired individuals that can autonomously detect various environmental hazards and provide real-time alerts. The specific objectives of this project are:

- To design and develop a smart stick that provides comprehensive safety support to visually impaired users.
- To detect static and dynamic obstacles such as walls, humans, and animals using an ultrasonic sensor.
- To detect water and ground-level hazards such as pits and uneven surfaces using a water sensor.
- To provide real-time alerts to the user through a buzzer, vibration motor, and voice module upon detection of obstacles or hazards.
- To implement an emergency communication system using GPS and GSM modules to send the user's location to a pre-saved contact.
- To understand the working of the Arduino UNO microcontroller and its interfacing with multiple sensor and output modules.
- To implement real-time decision-making logic based on multi-sensor input data.
- To design the system to be portable, low-cost, and power-efficient for practical everyday use.
- To gain hands-on knowledge of embedded systems programming, sensor integration, and circuit design.
- To evaluate the system's performance in different real-world environments including roads, indoor spaces, and crowded public areas.
- To contribute a practical and educational model that can be further enhanced with advanced technologies such as AI-powered object recognition.

4. METHODOLOGY

4.1 Requirement Analysis

- Identifying the safety needs of visually impaired individuals in different environments.
- Understanding the limitations of existing assistive tools such as traditional white canes.
- Defining the objectives of the proposed smart stick system including obstacle detection, hazard sensing, and emergency communication.
- Studying the working principles of ultrasonic sensors, water sensors, GPS, and GSM modules.

4.2 Component Selection

- Selecting Arduino UNO as the central microcontroller for processing sensor data and controlling outputs.
- Choosing HC-SR04 ultrasonic sensor for real-time obstacle detection.
- Using a water sensor to detect water and pit hazards on the ground.
- Integrating NEO-6M GPS module and SIM800L/SIM900 GSM module for emergency location sharing.
- Using a buzzer and vibration motor for tactile and audio alerts.
- Selecting a battery pack for portable power supply and jumper wires for circuit connections.

4.3 System Design

- Planning the overall structure of the smart stick to accommodate all components compactly.
- Deciding the placement of the ultrasonic sensor at the front of the stick for forward obstacle detection.
- Positioning the water sensor at the lower tip of the stick for ground-level hazard detection.
- Designing the circuit connection flow between the Arduino, sensors, GSM/GPS modules, and output devices.

4.4 Programming

- Writing the control code using the Arduino IDE in the Embedded C programming language.
- Implementing the logic for reading distance data from the ultrasonic sensor (HC-SR04).
- Programming alert conditions: activating the buzzer/vibration motor when an obstacle is detected within a set threshold.
- Implementing water detection logic to trigger a separate alert when the water sensor detects moisture.
- Programming the GPS module to retrieve the user's current location coordinates.
- Implementing the GSM module to send an SMS with the GPS coordinates to the pre-saved emergency contact.

ArduinoProgramCode:

```
// Pin definitions
const int trigPin = 9;
const int echoPin = 10;
const int buzzer = 11;

long duration;
int distance;

void setup() {
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
  pinMode(buzzer, OUTPUT);
  Serial.begin(9600);
}

void loop() {
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);

  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);

  duration = pulseIn(echoPin, HIGH);
  distance = duration * 0.034 / 2;

  if (distance <= 5) {
    digitalWrite(buzzer, HIGH);
    delay(200);
    digitalWrite(buzzer, LOW);
    delay(200);
  } else {
    digitalWrite(buzzer, LOW);
  }

  Serial.print("Distance: ");
  Serial.println(distance);

  delay(100);
}
```

4.5 Working Process

- The Arduino continuously reads data from the ultrasonic sensor and water sensor.
- If an obstacle is detected within a preset distance threshold, the buzzer/vibration motor is activated to alert the user.
- If the water sensor detects moisture, a separate alert is triggered via voice module.
- In an emergency, the user activates the GPS/GSM feature; the system retrieves the current GPS coordinates and sends an SMS to the saved contact.
- The system returns to normal monitoring mode after the alert condition is cleared.

4.6 Implementation

- Final assembly of all components on the stick body with secure wiring.
- Uploading the tested and verified code to the Arduino UNO board using the Arduino IDE.
- Conducting functional testing of each module individually and then as an integrated system.
- Ensuring smooth and reliable operation across different test scenarios.

5. TOOLS USED

The project utilizes both hardware components and software tools as described below.

5.1 Hardware Components

S.No	Component	Purpose
1	Arduino UNO / Nano	Main microcontroller; processes sensor data and controls all outputs.
2	Ultrasonic Sensor (HC-SR04)	Detects obstacles in the path of the user by measuring distance.
3	Water Sensor	Detects water on the ground and alerts the user to avoid slipping.
4	GPS Module (NEO-6M)	Retrieves current location coordinates for emergency use.
5	GSM Module (SIM800L/SIM900)	Sends SMS with GPS location to a pre-saved emergency contact.
6	Buzzer / Vibration Motor	Provides audio and tactile alerts upon obstacle or hazard detection.
7	Battery Pack	Provides portable power supply to the entire system.
8	Jumper Wires & Stick Body	Electrical connections and physical housing of the system.

5.2 Software Tools

Arduino IDE: The primary development environment used for writing, compiling, and uploading the embedded C code to the Arduino UNO board. It also provides the Serial Monitor for real-time debugging of sensor values.

Python (Optional): Used for additional data logging or processing tasks where required during testing and analysis phases.

Serial Monitor: A built-in tool in the Arduino IDE used to monitor real-time sensor readings and debug system behavior during development and testing.

6. SYSTEM ARCHITECTURE

6.1 Block Diagram

The block diagram of the Multifunctional Smart Stick illustrates the overall system architecture. The Arduino UNO acts as the central controller, receiving input from the ultrasonic sensor, the water sensor, and the GPS module. Based on the processed data, it outputs signals to the buzzer, vibration motor, and voice module for alerts, and to the GSM module for emergency communication.

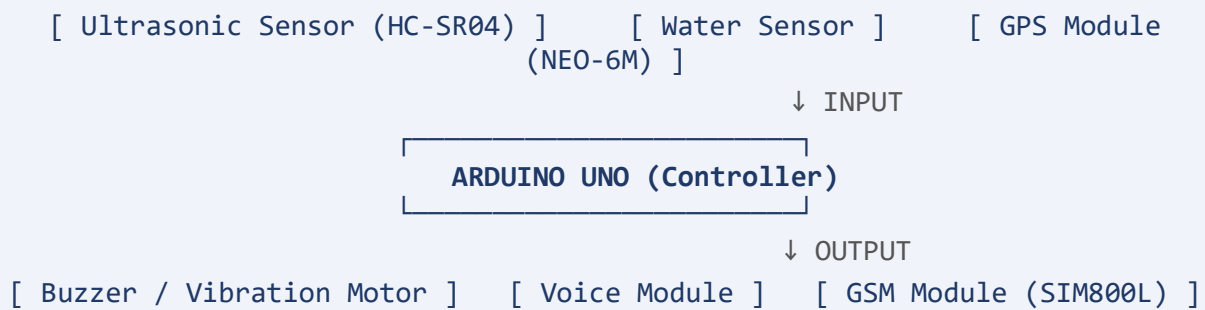
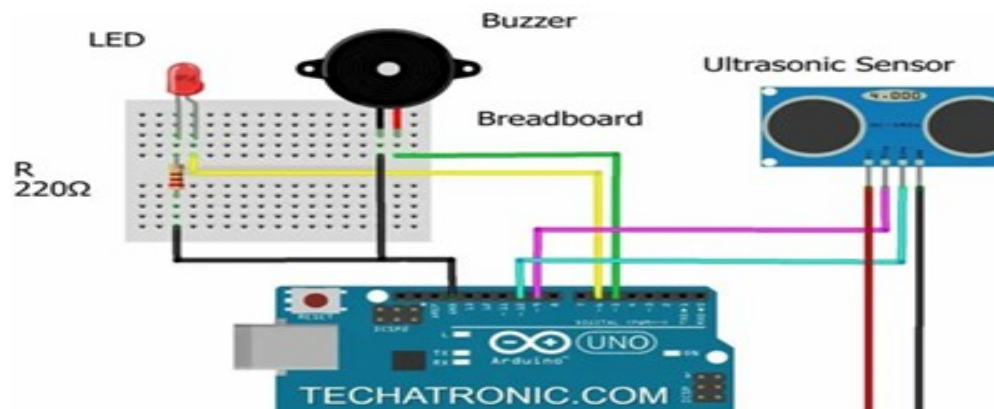


Fig 6.1: Block Diagram of Multifunctional Smart Stick

6.2 Circuit Diagram Overview

Circuit Diagram



7. WORKING PRINCIPLE

The working principle of the Multifunctional Smart Stick is based on continuously monitoring the environment through multiple sensors and responding to detected hazards with appropriate alerts. The system uses an ultrasonic sensor for obstacle detection, a water sensor for ground hazard detection, and GPS/GSM modules for emergency communication.

When the system is powered on, the Arduino UNO begins executing the control program. The ultrasonic sensor continuously emits ultrasonic waves and measures the time taken for the reflected signal to return. Based on this time measurement, the distance to any nearby obstacle is calculated. If this distance falls below a defined safe threshold, the Arduino activates the buzzer and vibration motor to immediately warn the user.

Simultaneously, the water sensor located at the lower end of the stick continuously monitors for the presence of water or moisture on the ground. If water is detected, the system triggers a separate alert via the buzzer or voice module, warning the user of the slipping hazard ahead.

In emergency situations, the user can trigger the GPS/GSM system. The GPS module retrieves the current geographic coordinates of the user, and the GSM module sends an SMS containing these coordinates to the pre-saved emergency contact number. This enables family members or caregivers to quickly locate the user in case of distress.

This integrated approach ensures that the user receives timely and specific warnings for different types of hazards, making navigation significantly safer and more independent than with conventional white canes.

8. RESULTS AND OBSERVATION

The system was tested under different real-world conditions and successfully demonstrated multi-hazard detection and alert functionality. The robot responded effectively by activating the appropriate alert mechanism for each type of detected hazard.

8.1 Observation Table

S.No	Distance (cm)	Detected Condition	System Action	Result
1	> 50	No obstacle / Clear path	Move freely; no alert	Safe navigation
2	20 – 50	Obstacle detected ahead	Buzzer activated / Vibration alert	User warned
3	< 20	Close obstacle / Imminent collision	Strong buzz + Voice alert	Collision avoided
4	Water detected	Water / Wet surface	Water alert via buzzer	Slip avoided
5	Emergency trigger	GPS/GSM activated	SMS sent with location	Family notified

Fig 8.1: Observation Table

9. APPLICATIONS

The Multifunctional Smart Stick has a wide range of applications that can significantly improve the quality of life for visually impaired individuals:

- For visually impaired people to walk safely and independently on roads and footpaths.
- In crowded public spaces such as railway stations, bus stands, markets, and shopping centers to detect moving people and obstacles.
- In hospitals, educational institutions, offices, and government buildings for safe indoor navigation.
- During nighttime or low-visibility conditions where traditional canes are insufficient.
- For detecting water puddles and uneven surfaces to prevent slipping and falling accidents.
- In emergency situations to automatically communicate the user's GPS location to family members or caregivers.
- In rehabilitation centers and old-age homes where residents may have limited vision.
- In research and education institutions as a demonstration of embedded systems and sensor integration for assistive technology.

10. CHALLENGES

During the development and testing of the Multifunctional Smart Stick, several technical and practical challenges were encountered:

One of the primary challenges was achieving accurate distance measurement using the ultrasonic sensor in real-world conditions. External noise, environmental factors such as wind, and highly absorbent or irregular surfaces sometimes affected the reliability of sensor readings. Calibration and testing were required to establish appropriate threshold values.

Integrating multiple communication modules, particularly the GPS and GSM modules, alongside the primary sensors proved to be challenging due to serial port conflicts and timing constraints on the Arduino UNO. Careful pin assignment and software serial communication configuration were necessary to resolve these issues.

Maintaining a stable and sufficient power supply for all components simultaneously was another challenge. The GPS, GSM, and motor driver modules have high peak current requirements, which can cause voltage drops that affect the stability of the Arduino and sensor readings. Proper power management was required to address this.

The GPS module required time to acquire satellite signals, particularly indoors or in areas with poor sky visibility. This limitation affects the reliability of the emergency location feature in such environments. Additionally, GSM connectivity depends on available network coverage.

Despite these challenges, systematic testing, debugging, and iterative improvements enabled the successful implementation of the system. The final prototype demonstrated reliable obstacle detection, hazard sensing, and emergency communication capabilities.

11. COMPARISON WITH EXISTING SYSTEMS

The proposed Multifunctional Smart Stick is compared with existing assistive systems to highlight its advantages and unique features:

Feature	Traditional White Cane / Basic Systems	Proposed Smart Stick
Obstacle Detection	Limited range, manual	Real-time via ultrasonic sensor
Water / Pit Detection	Not available	Available via water sensor
Dynamic Object Detection	Not available	Detected via ultrasonic sensor
Emergency Communication	Not available	GPS + GSM SMS alert
Alert Type	Physical cane contact	Buzzer, vibration & voice
Cost	Low	Low-cost, affordable
Portability	High	High, lightweight design

Fig 11.1: Comparison with Existing Systems

12. ADVANTAGES AND LIMITATIONS

12.1 Advantages

- Provides comprehensive safety through multi-sensor integration for obstacle, water, and dynamic object detection.
- Real-time obstacle detection ensures immediate alerts, significantly reducing the risk of accidents.
- Emergency GPS/GSM communication enables rapid location sharing with family members in critical situations.
- Compact, portable, and lightweight design makes it suitable for everyday use.
- Low cost and uses readily available components, making it accessible and affordable.
- Easy to use and does not require technical expertise from the end user.
- Serves as a strong educational model for embedded systems and assistive technology development.

12.2 Limitations

- Cannot detect very small, transparent, or highly sound-absorbent objects with the ultrasonic sensor.
- GPS module performance is reduced indoors or in urban areas with poor satellite visibility.
- The system requires periodic battery charging for continuous use.
- Performance may degrade under extreme weather conditions such as heavy rain or high humidity.
- The system lacks advanced artificial intelligence for complex object recognition or scene understanding.
- GSM-based emergency communication requires mobile network coverage, which may not be available in all areas.

13. CONCLUSION

The Multifunctional Smart Stick for Visually Impaired with Obstacle Detection was successfully designed and implemented using an Arduino UNO microcontroller, HC-SR04 ultrasonic sensor, water sensor, GPS module, GSM module, buzzer, and vibration motor. The system is capable of detecting obstacles, water hazards, and ground-level dangers in real time and provides immediate alerts through audio and tactile feedback.

The emergency communication feature, enabled by GPS and GSM modules, adds a critical safety layer by allowing the user's location to be shared with family members instantly. This significantly enhances the independence and safety of visually impaired users in public and private environments.

Through this project, valuable practical knowledge was gained in embedded systems programming, sensor interfacing, circuit design, and hardware integration. The project effectively demonstrates how affordable and accessible technologies can be combined to create high-impact solutions for real-world challenges.

Although the system performs well under tested conditions, there is room for further improvement. Future enhancements could include AI-powered object recognition using camera modules, Bluetooth or Wi-Fi connectivity for smartphone integration, and machine learning algorithms for improved decision-making. With such advancements, the system could evolve into a highly intelligent and comprehensive assistive device.

In conclusion, this project successfully proves that a multifunctional smart stick can be effectively built using simple and affordable components. It demonstrates the power of embedded systems in addressing real-world human challenges and lays a strong foundation for future development in the field of assistive technology.

14. FUTURE SCOPE

The Multifunctional Smart Stick has been successfully implemented, and several promising avenues exist for future enhancement:

- Integration of AI-powered object recognition using camera modules to identify and differentiate between different types of obstacles and hazards.
- Addition of Bluetooth or Wi-Fi connectivity to pair the stick with a smartphone application for additional navigation assistance and real-time monitoring.
- Implementation of machine learning algorithms to enable the system to learn from the user's environment and improve navigation decisions over time.
- Incorporation of a voice recognition module to allow users to interact with the system through voice commands.
- Development of a cloud-based monitoring dashboard for caregivers to track the user's location and activity in real time.
- Integration of temperature, humidity, and air quality sensors to provide additional environmental awareness.
- Use of solar charging panels to improve the battery life and reduce the dependency on manual charging.
- Deployment of advanced motor actuators within the stick for haptic feedback to communicate directional guidance to the user.

Overall, the future scope of this project is vast. With progressive enhancements in sensor technology, artificial intelligence, and wireless communication, the Multifunctional Smart Stick can be transformed into a fully intelligent autonomous assistive device capable of addressing a wide range of real-world navigation challenges for visually impaired individuals.

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